



# Designing clinically useful psychopharmacological trials: challenges and ways forward

Astrid Chevanche, Philippe Ravaud, Victoria Cornelius, Evan Mayo-Wilson, Toshi A Furukawa

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Université Paris Cité, CRESS,  
INSERM, INRAE, Paris, France

(A Chevanche PhD,

Prof P Ravaud PhD); Centre

d'Epidémiologie Clinique,

Hôpital Hôtel-Dieu, AP-HP,

Paris, France (A Chevanche,

Prof P Ravaud); Imperial Clinical

Trials Unit, School of Public

Health, Imperial College

London, London, UK

(V Cornelius PhD); Department

of Epidemiology and

Biostatistics, Indiana

University School of Public

Health-Bloomington,

Bloomington, IN, USA

(E Mayo-Wilson DPhil);

Department of Health

Promotion and Human

Behavior and Department of

Clinical Epidemiology, School

of Public Health, Kyoto

University Graduate School of

Medicine, Kyoto, Japan

(Prof T Furukawa PhD)

Correspondence to:

Dr Astrid Chevanche, Centre

d'Epidémiologie Clinique,

Hôpital Hôtel-Dieu, AP-HP,

75004 Paris, France

[astrid.chevanche@gmail.com](mailto:astrid.chevanche@gmail.com)

The clinical guidelines that underpin the use of drugs for mental disorders are informed by evidence from randomised controlled trials (RCTs). RCTs are performed to obtain marketing authorisation from regulators. The methods used in these RCTs could be appropriate for early phases of drug development because they identify drugs with important harms and drugs that are efficacious for specific health problems and populations. RCTs done before marketing authorisation do not tend to address clinical questions that concern the effectiveness of a drug in heterogeneous and comorbid populations, the optimisation of drug sequencing and discontinuation, or the comparative benefits and harms of different drugs that could be used for the same health problem. This Review proposes an overview of some shortcomings of RCTs, at an individual level and at the whole portfolio level, and identifies some methods in planning, conducting, and carrying out analyses in RCTs that could enhance their ability to support therapeutic decisions. These suggestions include: identifying patient-important questions to be investigated by psychopharmacological RCTs; embedding pragmatic RCTs within clinical practice to improve generalisability to target populations; collecting evidence about drugs in overlooked populations; developing methods to facilitate the recruitment of patients with mental disorders and to reduce the number of patients who drop out, using specific methods; using core outcome sets to standardise the assessment of benefits and harms; and recording systematically serious objective outcomes, such as suicide or hospitalisation, to be evaluated in meta-analyses. This work is a call to address questions relevant to patients using diverse design of RCTs, thus contributing to the development of a patient-centred, evidence-based psychiatry.

## Introduction

Mental disorders have an estimated annual prevalence of 20% and a lifetime prevalence of 33%.<sup>1</sup> Mental disorders often begin in early adulthood, and are among the top three causes of disability worldwide.<sup>2</sup> Psychopharmacological treatments (drugs) for these disorders are increasing, with antidepressant consumption doubling since 2000 in the Organisation for Economic Cooperation and Development countries.<sup>3,4</sup>

Since the 1980s, multicentre, double-blind, randomised controlled trials (RCTs) have been the reference method for obtaining marketing authorisation for new pharmacological interventions from regulatory agencies such as the US Food and Drug Administration (FDA) and the European Medicines Agency (EMA). Between 2007 and 2018, 11 168 mental health trials were registered on ClinicalTrials.gov, of which 6302 (56%) were done in the USA.<sup>5</sup> Among these 6302 trials, 5004 (79%) were randomised and 1942 (31%) were double-blinded. Furthermore, 2189 (35%) trials investigated just drugs and 581 (9%) investigated drugs and other interventions (eg, behavioural interventions).<sup>5</sup> Of the 2940 (47%) trials using US FDA-defined phases, only 632 (21%) were postmarketing trials (phase 4 trials), whereas 2308 (79%) were premarketing trials (phase 1 to phase 3 trials).<sup>5</sup> Clinical guidelines often rely on these phase 1–3 trials, which might have low generalisability for, or might overlook, outcomes that matter to patients.<sup>6–8</sup>

A patient-centred, evidence-based psychiatry requires RCTs that are performed using methods that facilitate the translation of their results into daily clinical routine. This research investment could enhance the lives of the millions of people with psychiatric disorders across the globe, and the lives of their families. This Review addresses the ability

of both individual RCTs and entire portfolios of RCTs to provide evidence to support therapeutic decisions made on the basis of outcomes that matter to patients. Important choices concern the harms and benefits of drugs, drug sequences, and how and when to introduce or discontinue drugs. We propose a broad list of possible methodological improvements and provide references for readers seeking out more details on specific topics (panel 1). Issues that are not specific to psychopharmacological RCTs, such as participant randomisation and dealing with missing data, are not addressed here.

## Improving the value of psychopharmacological RCTs by addressing questions that matter to patients and clinicians

Currently, there is a disconnect between daily clinical problems reported by patients and clinicians and the available evidence in psychiatry. We propose three areas for improving the values of RCTs.

Firstly, although many mental health problems are chronic and might necessitate drugs for many years, the clinical guidelines underpinning these prescriptions are based on results from RCTs with insufficient follow-up duration. For example, RCTs of antidepressants typically measure outcomes at approximately 2 months. The mean duration of the 522 RCTs (representing 116 477 patients) included in the largest network meta-analysis was 8 weeks (IQR 6–8). Of these 522 trials, 502 RCTs (96·0%; representing 111 544 patients [96·2%]) lasted less than 6 months, 18 RCTs (3·4%, representing 4933 patients [4·2%]) lasted 6 months, and the remaining two RCTs (0·4% representing 776 patients [0·7%]) lasted 9 months.<sup>9</sup> In 2021, a network meta-analysis of the maintenance of treatment for depression included only